Superposition and interference

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Add signed displacements and distinguish interference from lost energy.

SUCCESS CRITERIA - I can...

1. Add wave displacements at the same place and time.
2. Recognize constructive and destructive interference.
3. Explain that pulses continue after crossing in a linear medium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “superposition” and “interference” to compare your model or result. Sentence starters are optional.

superposition: adding displacements or fields at the same place and time

interference: the pattern produced by overlapping waves

displacement: change from initial position to final position, with direction

01 Bring back an earlier idea

An upward rope pulse is 3 cm high before reaching an ideal free end. Is the reflected displacement upright or inverted?
Sketch it.

Word help: ideal = using stated simplifying conditions

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $y_{resultant} = y_1 + y_2$.
Add signed displacements at the same place and time. Local cancellation does not destroy all wave energy.

Worked reasoning: At one point, two pulses have displacements +3 cm and -1 cm. Superposition gives +2 cm there. Add signed displacements, not their absolute values.

Your turn: Draw the two component displacements and the resultant at that point.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

Two upward pulses overlap.
Sketch the combined displacement. Then sketch equal upward/downward pulses overlapping.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Build the solution

At one point displacements are +3 cm and -1 cm.
Find total. Repeat for +2 cm and -2 cm.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Two rope pulses momentarily cancel. A student says the pulses are gone forever. What happens after they pass in an ideal linear medium?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A system receives 60 J heat and does 40 J work on its surroundings.
Find ΔU with $\Delta U = Q - W_{by}$.

Word help: surroundings = everything outside the chosen system

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Add displacements -4 mm and +7 mm.
Name whether they reinforce or partly cancel.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Construct three different pairs of displacements with resultant zero. Can the waves still move after that instant?

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3